Summary. Let V be a finite dim. vect. sp. over K. Then:

1) Every finite list of vectors that spans V contains a sublist which is a basis of V.

$(v_1, \dots, v_n \rightsquigarrow v_{i_1}, \dots, v_{i_d}\ ;\ 1 \leq i_1 < i_2 < \dots < i_d \leq n,\ d = \dim V.)$

2) Every lin. indep. list can be extended to a basis of V.

$(u_1, \dots, u_m \rightsquigarrow u_1, \dots, u_m, w_1, \dots, w_{d-m}\ ;\ d = \dim V.)$

Thm 1. Let V be a vect. space of $\dim V = n \in \mathbb{Z}_{\geq 0}$. Let $v_1, \dots, v_n \in V$. Then the following statements are equivalent:

1) $v_1, \dots, v_n$ are lin. indep.
2) $v_1, \dots, v_n$ generate V, i.e. $Sp(v_1, \dots, v_n) = V$.
3) $v_1, \dots, v_n$ is a basis for V.

Proof. $1 \Rightarrow 3$: By the summary above we can extend the list $v_1, \dots, v_n$ to a basis. But $v_1, \dots, v_n$ has already length $n = \dim V$. $\Rightarrow$ $v_1, \dots, v_n$ is already a basis.

$3 \Rightarrow 1 \,\&\, 2$: This is part of the def. of basis.

$2 \Rightarrow 3$: If $v_1, \dots, v_n$ generate V, then by the summary above a sublist of $v_1, \dots, v_n$ is a basis. But $v_1, \dots, v_n$ contains $n = \dim V$ elements. $\Rightarrow$ This is already a basis. ∎

Thm 2. Suppose $n = \dim V < \infty$. Let $v_1, \dots, v_k$ be a list of vectors in V. Then:

1) If $k < n$, the list $v_1, \dots, v_k$ can NOT span V.

2) If $k > n$, the list $v_1, \dots, v_k$ can NOT be lin. indep.

Proof. 1) Assume that $k < n$.

Suppose by contradiction that $sp(v_1, \dots, v_k) = V$.

By the summary above we can delete some of the vectors from the list $v_1, \dots, v_k$ and get a basis $v_{i_1}, \dots, v_{i_{k'}}$ of V. $\Rightarrow \dim V = k' \leq k < n$. Contradiction.

2) Assume $k > n$. Suppose by contradiction that $v_1, \dots, v_k$ are lin. indep. $\Rightarrow$ we can extend this list to a basis of V. $\Rightarrow \dim V \geq k > n$. Contradiction. ∎

Recall from prev. lectures:

Lemma E. Let $w_1, \dots, w_\ell \in V$ and assume that $w_j \notin sp(w_1, \dots, w_{j-1})$ $\forall$ $1 \leq j \leq \ell$. Then the list $w_1, \dots, w_\ell$ is linear independent.

Prop. Let V be a finite dim. vector space over K.
Then every subspace $\mathcal{U} \subseteq V$ is also finite dim.
Moreover $\dim \mathcal{U} \leq \dim V$ with equality iff $\mathcal{U} = V$.

Proof. Denote $n := \dim V$. Let $\mathcal{U} \subseteq V$ be a subspace.
We'll first show that $\mathcal{U}$ has a finite basis.
If $\mathcal{U} = \{0\}$ then $\emptyset$ is the basis of $\mathcal{U}$ and we are done. So assume now $\mathcal{U} \neq \{0\}$. Take $v_1 \in \mathcal{U} \setminus \{0\}$.
If $\mathcal{U} = sp(v_1)$ we get a basis for $\mathcal{U}$.
If $\mathcal{U} \neq sp(v_1)$, then take $v_2 \in \mathcal{U} \setminus sp(v_1)$. By Lemma E v_1, v_2 are lin. indep.
We continue this process and after j steps we obtain a list $v_1, \dots, v_j \in \mathcal{U}$ that is lin. indep.
If $sp(v_1, \dots, v_j) = \mathcal{U}$ we are done. Otherwise we choose $v_{j+1} \in \mathcal{U} \setminus sp(v_1, \dots, v_j)$. Again by Lemma E $v_1, \dots, v_j, v_{j+1}$ are lin. indep. However, in V a list of lin. indep. vectors can have length n at most.
$\Rightarrow$ The procedure described above must end after not more than n step, say $k \leq n$ steps.
And we obtain a list $v_1, \dots, v_k \in \mathcal{U}$, $k \leq n$, of lin. indep. vectors s.t. $sp(v_1, \dots, v_k) = \mathcal{U}$.
By def $v_1, \dots, v_k$ is a basis for $\mathcal{U}$. $\Rightarrow \dim \mathcal{U} = k \leq n = \dim V$.

Now, if $\mathcal{U} = V$ then clearly $\dim \mathcal{U} = \dim V$.

Conversely, suppose $\dim \mathcal{U} = \dim V$. $\Rightarrow k = n$.

The list of vectors $v_1, \dots, v_n \in \mathcal{U}$ is a basis for $\mathcal{U}$, hence are lin. indep. also when considered in V.

By Thm 1., $v_1, \dots, v_n$ form a basis for V.

$\Rightarrow \quad \mathcal{U} = sp(v_1, \dots, v_n) = V.$ ∎